

shown for a frog tadpole in **Figure 47.22a**. The right-left axis is largely symmetrical, as the two sides are roughly mirror images. When and how are these three axes established? We'll begin answering this question by considering the frog.

Axis Formation in the Frog

In the frog, the future position of the anterior-posterior axis is determined during oogenesis. Asymmetry in the egg is apparent in the formation of two distinct hemispheres: Dark melanin granules are embedded in the cortex of the animal hemisphere, whereas a yellow yolk fills the vegetal hemisphere. This animal-vegetal asymmetry dictates where the anterior-posterior axis forms in the embryo. Note, however, that the anterior-posterior and animal-vegetal axes are not the same; that is, the head of the embryo does not coincide with the animal pole.

Surprisingly, the dorsal-ventral axis of the frog embryo is determined at random. Specifically, wherever the sperm enters in the animal hemisphere determines where the dorsal-ventral axis forms. Once the sperm and egg have fused, the egg surface—the plasma membrane and associated cortex—rotates with respect to the inner cytoplasm, a movement called *cortical rotation*. From the perspective of the animal pole, this rotation is always toward the point of sperm entry (**Figure 47.22b**). The resulting interactions between molecules in the vegetal cortex and in the inner cytoplasm of the animal hemisphere activate regulatory proteins. Once activated, these proteins direct expression of one set of genes in dorsal regions and another set of genes in ventral regions.

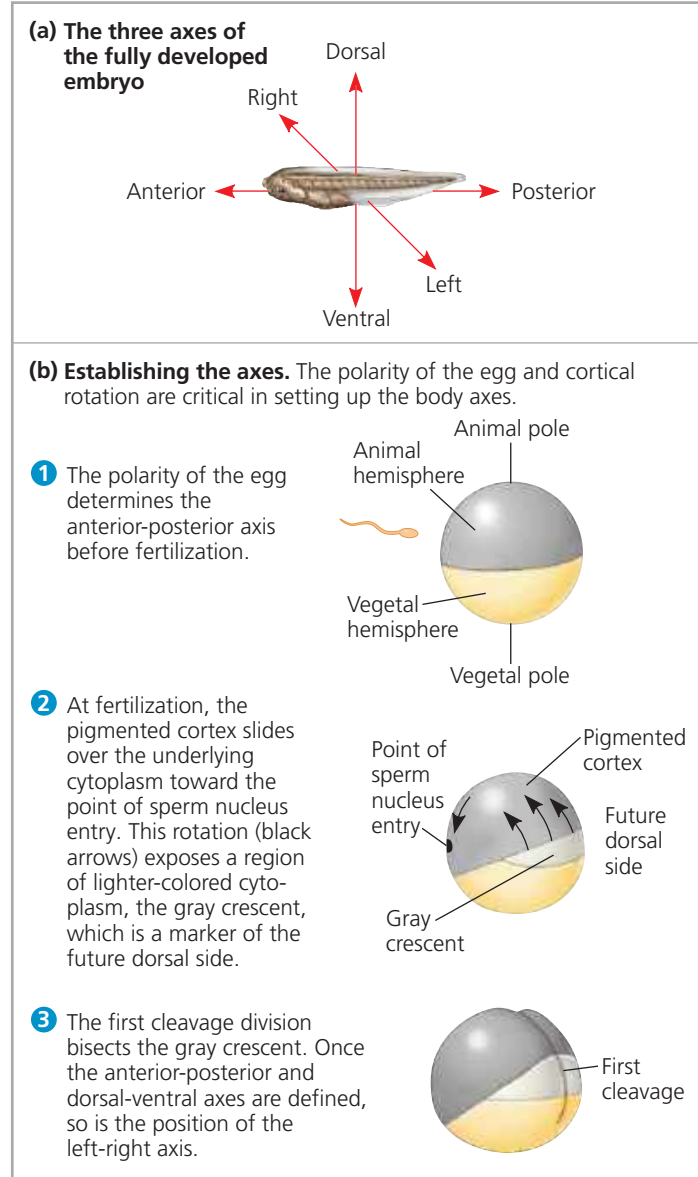
Axis Formation in Birds, Mammals, and Insects

It turns out that there are many different processes by which animal embryos establish their body axes. In mammals, the sperm appears to contribute to axis formation, but not in the same manner as in frogs. In particular, the orientation of the egg and sperm nuclei before they fuse influences the location of the first cleavage plane. In chicks, the anterior-posterior axis is established by the pull of gravity during the time when the soon-to-be-laid egg is traveling down the hen's oviduct. In zebrafish, signals within the embryo gradually establish the anterior-posterior axis over the course of a day. Still other mechanisms occur in insects, where gradients of active transcription factors across the body establish both the anterior-posterior and dorsal-ventral axes (see Concept 18.4).

Once the anterior-posterior and dorsal-ventral axes are established, the position of the left-right axis is fixed. Nevertheless, specific molecular mechanisms must establish which side is left and which is right. In vertebrates, there are marked left-right differences in the location of internal organs as well as in the organization and structure of the heart and brain. Recent research has revealed a key role for cilia in setting up this left-right asymmetry, as we will discuss toward the end of this chapter.



▼ **Figure 47.22 The body axes and their establishment in an amphibian.** All three axes are established before the zygote begins to undergo cleavage.



WHAT IF? When researchers allowed normal cortical rotation to occur and then forced the opposite rotation, the result was a two-headed embryo. How might you explain this finding, thinking about how cortical rotation influences body axis formation?

Restricting Developmental Potential

Earlier we described determination in terms of commitment to a particular cell fate. The fertilized egg gives rise to all cell fates. How long during development do cells retain this ability? The German zoologist Hans Spemann addressed this question in 1938. By manipulating embryos to perturb normal development and then examining cell fate after the manipulation, he was able to assay a cell's *developmental potential*, the range of structures to which it can give rise